

HW3 Q 7

MDS. Prove that classical (metric) Multi-Dimensional Scaling (MDS) with Euclidean distances is equivalent to PCA.

MDS Distance b/w every pair of samples (i, j) ; μ : mean of sample
 $B_{ij} = \langle x_i - \mu, x_j - \mu \rangle \quad \forall i, j = 1, 2, \dots, n$

$$H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^T, \quad H: \text{centering matrix}$$

The distance between x_i and x_j is:

$$d_{ij}^2 = \|x_i - x_j\|^2 = (x_i - x_j)^T (x_i - x_j) = x_i^T x_i + x_j^T x_j - 2x_i^T x_j$$

Let K be the inner product matrix

$$K = X^T X, \quad \text{with } k = \text{diag}(K_{ii}) \in \mathbb{R}^n, \text{ So,}$$

$$D = (d_{ij}^2) = k \cdot \mathbf{1}\mathbf{1}^T + \mathbf{1} \cdot k^T - 2K$$

$$\left\{ \tilde{x}_i = x_i - \hat{\mu}_n = x_i - \frac{1}{n} \cdot X \cdot \mathbf{1} \right\}$$

$$\tilde{X} = X - \frac{1}{n} X \cdot \mathbf{1} \cdot \mathbf{1}^T$$

Thus,

$$\tilde{K} \triangleq \tilde{X}^T \tilde{X} = \left(X - \frac{1}{n} X \cdot \mathbf{1} \cdot \mathbf{1}^T \right)^T \left(X - \frac{1}{n} X \cdot \mathbf{1} \cdot \mathbf{1}^T \right)$$

$$= K - \frac{1}{n} K \cdot \mathbf{1} \cdot \mathbf{1}^T - \frac{1}{n} \mathbf{1} \mathbf{1}^T K + \frac{1}{n^2} \mathbf{1} \cdot \mathbf{1}^T \cdot K \cdot \mathbf{1} \cdot \mathbf{1}^T$$

$$\text{Let } B = -\frac{1}{2} H \cdot D \cdot H^T$$

$$\text{where } H = I - \frac{1}{n} \mathbf{1} \cdot \mathbf{1}^T \quad (H: \text{centering matrix})$$

So,

$$B = -\frac{1}{2} H \cdot (k \cdot 1^T + 1 \cdot k^T - 2K) \cdot H^T$$

Since $k \cdot 1^T \cdot H^T = k \cdot 1 \left(I - \frac{1}{n} \cdot 11^T \right)$
 $= k \cdot 1 - k \left(\frac{1^T \cdot 1}{n} \right) \cdot 1 = 0$

we have

$$H \cdot k \cdot 1 \cdot H^T = H \cdot 1 \cdot k^T \cdot H^T = 0$$

Therefore:

$$B = H \cdot K \cdot H^T = \left(I - \frac{1}{n} \cdot 11^T \right) \cdot K \cdot \left(I - \frac{1}{n} \cdot 11^T \right)$$

that is: $= K - \frac{1}{n} \cdot 11^T \cdot K - \frac{1}{n} \cdot K \cdot 11^T + \frac{1}{n^2} \cdot 1(1^T \cdot K \cdot 1)1^T$

$$B = \frac{1}{2} H \cdot D \cdot H^T = \tilde{X}^T \tilde{X}$$

$$\tilde{X} = H \cdot X$$

$X \in \mathbb{R}^{n \times m} \quad X \in \mathbb{R}^{m \times n}$

$$* B = H X X^T H$$

$$* B = \tilde{X}^T \tilde{X} = E \Lambda E^T$$

$$\tilde{X}^{mos} = E_m \Lambda_m^{1/2}$$

PCA

$$x_1, x_2, x_3, \dots, x_n \in \mathbb{R}^p, \quad X = [x_1, x_2, x_3, \dots, x_n]^{p \times n}$$

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \cdot X \cdot 1, \quad 1 = (1, 1, 1, \dots, 1)^T \in \mathbb{R}^n$$

$$\tilde{X} = X - \frac{1}{n} X \cdot 1 \cdot 1^T; \quad \tilde{x}_i = x_i - \hat{\mu}_n$$

$$\hat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu}_n) (x_i - \hat{\mu}_n)^T = \frac{1}{n} (\tilde{X} \cdot \tilde{X}^T)$$

$$H = I - \frac{1}{n} 11^T$$

$$* \tilde{X} = H \cdot X = USV^T$$

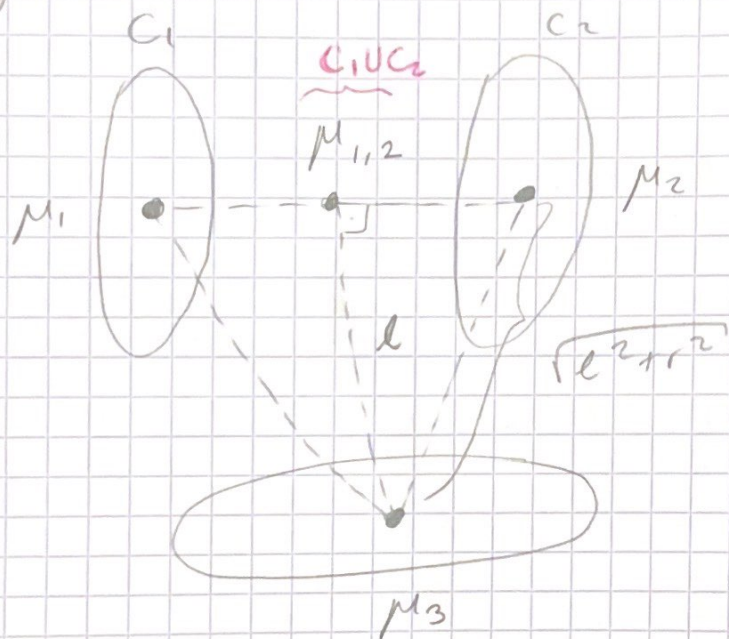
$$\tilde{X}^{PCA} = U_m S_m$$

$$E \Lambda E^T = B = H X X^T H = USV^T V S U^T = US^2 U^T$$

$$* \tilde{X}^{mos} = E_m \Lambda_m^{1/2} = U_m S_m = \tilde{X}^{PCA}$$

HW3 Q8

Hierarchical Clustering. Prove that centroid linkage in hierarchical clustering can result in inversions in the dendrogram.



Let

$$n_1 = n_2 = n_3 = n$$

$$n_3 = 0$$

$$S_i = \sum_{j \in C_i} X_j$$

$$r = \frac{1}{2} d(C_1, C_2) = \frac{1}{2} \left| \frac{1}{n} S_1 - \frac{1}{n} S_2 \right|$$

$$l = d((C_1 \cup C_2), C_3) = \left| \frac{1}{2n} (S_1 + S_2) - \frac{1}{n} S_3 \right|$$

$$\textcircled{1} \quad 2r < \sqrt{l^2 + r^2}$$

$$\text{let } \sqrt{3} |S_1 - S_2| < |S_1 + S_2| < 2 |S_1 - S_2| \quad \left\{ \begin{array}{l} \textcircled{*} \\ 2r < \sqrt{l^2 + r^2} \end{array} \right\}$$

$$\text{So } l = \left| \frac{1}{2n} (S_1 + S_2) \right| > r\sqrt{3} \Rightarrow r\sqrt{3} < l$$

$$\Rightarrow 3r^2 < l^2 \Rightarrow 2r < \sqrt{l^2 + r^2}$$

$$② \quad d < 2r$$

$$l = \frac{1}{2} | (s_1 + s_2) | < \frac{1}{n} | s_1 - s_2 | = 2r$$

$$\Rightarrow l < 2r$$

$$\Rightarrow l < 2r < \sqrt{l^2 + r^2}$$

$$③ \quad d(c_1 \cup c_2, c_3) < d(c_1, c_2)$$

This inversion occurs because centroid linkage relies solely on the distance between the centroids, which may not always accurately reflect the overall similarity between clusters. This is why centroid linkage can lead to dendrogram inversions.

HW 3. Q. 9

Spectral Clustering. Consider an undirected graph with L non negative edge weights $w_{i,j}$, and Graph Laplacian. Suppose there are m connected components, $A_1, A_2, \dots, A_m$ in the graph. Show that there are m eigenvectors of L that have an associated eigenvalue of zero, and that the indicator vectors of these components, $\mathbb{1}_{A_1}, \mathbb{1}_{A_2}, \dots, \mathbb{1}_{A_m}$ span the zero eigenspace.

(nodes edges)

$$G = (V, E) \quad V = \{1, 2, 3, \dots, N\} \quad E \subseteq V \times V,$$

$$\left[\begin{array}{l} \text{Let } i, \text{ and } j \text{ be on edge} \\ \text{in the graph, where;} \\ (i, j) \in E \iff \\ i, j \text{ are connected} \end{array} \right] \quad \left[\begin{array}{l} V = \{v_1, v_2, \dots, v_m\} \\ E = \{e_1, e_2, \dots, e_m\} \end{array} \right]$$

$$A \in \mathbb{R}^{N \times N}, \quad A_{i,j} \neq 0 \text{ iff } (i,j) \in E \quad (A: \text{adjacency matrix})$$

(nonzero values when we have edges in the graph)

$$d \in \mathbb{R}^N, \quad d_i = \sum_{j=1}^N A_{i,j}$$

d : degree vector, d_i

$$L = \text{diag}(d) - A$$

Let's have some non-zero vector V ;

$$V \neq 0, \quad LV = 0$$

L : eigenvector 0 : eigenvalue

$$V^T L V = \frac{1}{2} \sum_{(i,j) \in E} (v_i - v_j)^2 = \frac{1}{2} \sum_{k=1}^m \sum_{(i,j) \in E_k} (v_i - v_j)^2 = 0$$

If V is constant on each connected component,

$$\text{then } V^T L V = 0$$

$$V = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad V^T L V = 0$$

$$(A_1: v_1^{(k)} = v_2^{(k)})$$

So $\{v_i\}_{i=1}^m = \{I_{A_i}\}_{i=1}^m$ $I_{A_i}^T; I_{A_j} = 0 \forall i \neq j$
The indicator vectors are all eigenvectors of L , with eigenvalue 0

Lets assume that there is some vector; $V \neq 0$

Let $v \neq 0$ s.t. $Lv = 0$ $v \neq I_{A_i} \forall i$, $V^T I_{A_i} = 0 \forall i$

$$\exists \gamma \in \mathbb{R}^m \text{ st } v = \sum_{i=1}^m \gamma_i I_{A_i}$$

$$\Rightarrow V^T I_{A_i} \neq 0 \Rightarrow \text{not possible}$$

This shows that there are only m eigenvectors of L , that have an associated eigenvalue of zero, because that is how many connected components there are

The number of eigenvalues are equal to the number of connected component in the graph.

HW3 - Q10

Gaussian Graphical Models Assume X is normally distributed with covariance Θ^{-1} . Then derive how the partial correlation between X_i and X_j , $\rho_{ij|\neq i,j}$ is related to Θ_{ij} and $\hat{\beta}_j^{(i)}$ where $\hat{\beta}_j^{(i)}$ is the minimizer of $\|X_i - X_{\neq i} \beta^{(i)}\|_2^2$.

X : N samples of gaussian data, l : two entries. C : ^{except i, j} all variable

$$X \sim N(\mu, \Sigma) \quad l = \{i, j\} \subseteq \{1, 2, \dots, p\} \quad C = \{1, 2, \dots, p\} \setminus l$$

$$\mu_{A|B} = \mu_A + \Sigma_{AB} \Sigma_{BB}^{-1} (X_B - \mu_B) \quad \text{-- conditional mean}$$

$$\Sigma_{A|B} = \Sigma_A - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA} \quad \text{-- conditional variance}$$

$$\left(\mu_{i|j,c} \stackrel{?}{=} \mu_{i|c} \quad \mu_{j|i,c} \stackrel{?}{=} \mu_{j|c} \right)$$

$$\mu_{i|j,c} = \mu_{i|c} + \left(\Sigma_{\{i,j\}c} \right)_{ij} \underbrace{\left(\Sigma_{\{i,j\}c} \right)_{ji}^{-1}}_{>0} \underbrace{(X_{j|c} - \mu_{j|c})}_{\text{nonzero}}$$

$$X_i \perp X_j | X_c \iff \mu_{i|j,c} = \mu_{i|c} \iff \left(\Sigma_{\{i,j\}c} \right)_{ij}^{-1} = 0$$

$$\iff \Sigma_{\{i,j\}c} \text{ is diagonal}$$

Schur complements: $\Theta_{AA}^{-1} = \Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA} = \Sigma_{A|B}$

$$-\Theta_{AA}^{-1} \Theta_{AB} = \Sigma_{AB} \Sigma_{BB}^{-1}$$

When

$$\Sigma_{\{i,j\}c} \text{ diagonal} \iff \Theta_{\{i,j\}c} \text{ diagonal} \iff \Theta_{\{i,j\}c} \text{ diagonal} \iff \Theta_{ij} = 0$$

$$* \left[X_i \perp X_j | X_c \iff \Theta_{ij} = 0 \right] *$$

$$\Sigma_{AB} = \Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA} \quad \text{and}$$

$$\Theta_{AA^{-1}} = \Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA} = \Sigma_{A|B}$$

So, $-\Theta_{AA^{-1}} \Theta_{AB} = \Sigma_{AB} \Sigma_{BB}^{-1} \quad A=i, B=\{1,2,\dots,p\} \mid \{i\} = "i"$

$$-\Theta_{ii^{-1}} \Theta_{\neq i, i} = \underbrace{\Sigma_{\neq i, \neq i}}_{\neq i} \underbrace{\Sigma_{\neq i, i}}_{\neq i} = \left(X_{\neq i}^T X_{\neq i} \right)^{-1} \left(X_{\neq i}^T X_i \right) = \hat{\beta}^{(i)}$$

$$\hat{\beta}^{(i)} = \underset{\beta}{\operatorname{argmin}} \|X_i - X_{\neq i} \beta\|_2^2 = \left(X_{\neq i}^T X_{\neq i} \right)^{-1} \left(X_{\neq i}^T X_i \right)$$

$$\beta_j^{(i)} = -\left(\Theta_{ii^{-1}} \Theta_{\neq i, i} \right)_j = -\frac{\Theta_{ij}}{\Theta_{ii}}$$

$$\Rightarrow \hat{\beta}_j^{(i)} = 0 \iff \Theta_{ij} = 0 \iff X_i \perp X_j \mid X_c$$

* So $\Theta_{ij} = 0$ then $X_i \perp X_j \mid X_c$ and $\hat{\beta}_j^{(i)}$ estimates i^{th} column of U *

Partial

$$\rho_{\{i,j\} \mid c} = \frac{\operatorname{cov}(e_i, e_j)}{\sqrt{\operatorname{cov}(e_i, e_i) \operatorname{cov}(e_j, e_j)}} = \frac{e_i^T e_j}{\sqrt{(e_i^T e_i)(e_j^T e_j)}}$$

$$e_i = X_i - X_c \hat{\beta}^{(i)} \quad e_j = X_j - X_c \hat{\beta}^{(j)}$$

$$\beta^{(i)} = \underset{\beta}{\operatorname{argmin}} \|X_i - X_c \beta\|_2^2 = \Sigma_{cc}^{-1} \Sigma_{ci}, \quad \tilde{\beta}^{(i)} = \Sigma_{cc}^{-1} \Sigma_{ci}$$

$$\begin{aligned} \text{cov}(e_i, e_j) &= \frac{1}{n} e_i^T e_j = \frac{1}{n} \left(X_i^T X_j - x_i X_c \Sigma_{cc}^{-1} \Sigma_{cj} - \right. \\ &\quad \left. - \Sigma_{ic} \Sigma_{cc}^{-1} X_c^T X_j + \Sigma_{ic} \Sigma_{cc}^{-1} X_c X_c \Sigma_{cc}^{-1} \Sigma_{cj} \right) \\ &= \Sigma_{ij} - \Sigma_{ic} \Sigma_{cc}^{-1} \Sigma_{cj} = \left(\Theta_{\{i,j\}\{i,j\}} \right)^{-1}_{ij} \end{aligned}$$

$$\Theta_{\{i,j\}\{i,j\}}^{-1} = \Sigma_{\{i,j\}\{i,j\}} - \Sigma_{\{i,j\}c} \Sigma_{cc}^{-1} \Sigma_{c\{i,j\}}$$

$$= \begin{bmatrix} \text{cov}(e_i, e_i) & \text{cov}(e_i, e_j) \\ \text{cov}(e_j, e_i) & \text{cov}(e_j, e_j) \end{bmatrix}$$

$$\Theta_{\{i,j\}\{i,j\}} = \frac{1}{\det \Theta_{\{i,j\}\{i,j\}}} \begin{bmatrix} \text{cov}(e_j, e_j) & -\text{cov}(e_j, e_i) \\ -\text{cov}(e_i, e_j) & \text{cov}(e_i, e_i) \end{bmatrix}$$

$$\begin{aligned} P_{\{i,j\} | c} &= \frac{-\det \Theta_{\{i,j\}\{i,j\}} \Theta_{ij}}{\sqrt{(\det \Theta_{\{i\}\{i\}}) \Theta_{ii} (\det \Theta_{\{j\}\{j\}}) \Theta_{jj}}} \\ &= \frac{-\Theta_{ij}}{\sqrt{\Theta_{ii} \Theta_{jj}}} \end{aligned}$$

$$* \left[P_{\{i,j\} | c} = 0 \iff \Theta_{ij} = 0 \iff X_i \perp X_j | X_c \iff \hat{\beta}_j^{(i)} = 0 \right] *$$